Daniel Engelberg

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Herzliya, Israel | Authorized to work in the US | Relocating to New York

SUMMARY

Engineer (B.Sc. Mechanical Engineering, Technion, robotics) who turns field data into work crews can act on: built a LiDAR pipeline that finds trees too close to power lines on the worst hot, iced and windy day, validated end to end against hand labels, plus a production WhatsApp AI platform that runs maintenance operations.

SKILLS

LiDAR and geometry: Point clouds (laspy, LAS), ground filtering, catenary fitting, RANSAC, DBSCAN, KD-trees, GeoJSON, NumPy, SciPy
Programming: Python, JavaScript, TypeScript, Node.js, SQL, MATLAB
Robotics: ROS, kinematics, path planning, control theory, simulation, signal processing, RL (PPO)
AI: LLM APIs (Claude, OpenAI, OpenRouter), structured output, AI agents, speech-to-text, MediaPipe
Languages: English (native), Hebrew (fluent)

EDUCATION

Technion – Israel Institute of Technology, Haifa, Israel Mar 2026
B.Sc. in Mechanical Engineering, Robotics specialization
Relevant coursework: Python, Robotics (Kinematics, Path Planning), Control Theory, Signal Processing, CAD, Mechanical Design, Manufacturing Processes, Fluid Mechanics, Solid Mechanics

PROJECTS

Tree-to-Line Clearance from LiDAR | *Python, NumPy, SciPy, scikit-learn, laspy* Oct 2026 – Nov 2026

- Built a pipeline that turns raw drone LiDAR of high-voltage lines into a vegetation work list: classifies every point with geometric rules (no trained model), rebuilds towers, spans and each conductor as a catenary (2–7 cm fit), and flags trees too close to the line
- Moved each conductor to 40 °C, ice load and a 500 N/m² wind with the change-of-state equation and measured every tree against the Swiss legal clearances, reporting tiered encroachment zones, fall-in hazard trees and wire it could not assess
- Scored end to end against hand labels on 7 tiles (389M points): on a held-out forest tile, 34 of 34 encroachment zones found, none false, all in the right tier; traced a second held-out tile's crash and fixed the 4 defects it exposed
- Shipped as a tested Python package (43 tests, CI on GitHub Actions) with CSV, GeoJSON and LAS outputs; live at lidar-clearance.vercel.app

Handy AI | *Python, Flask, PostgreSQL, Claude / LLM APIs, WhatsApp Cloud API* Oct 2026 – Present

- Built a WhatsApp AI maintenance coordinator for property managers: tenants report issues by text, photo or voice note, and the system runs intake, collects contractor quotes, schedules visits and requests payment
- Engineered a Postgres-backed job queue (retries, dead-letter, per-conversation ordering, deduplication) and a 13-state ticket lifecycle with a full audit trail
- Wrote 350+ automated tests across 18,000+ lines of Python; deployed the web service and background worker on Render

Hotel Robot Operating System

- Built a hotel operations system that dispatches delivery and logistics robots, coordinates elevator rides and takes requests from the property management system (PMS)

EXPERIENCE

FlowTransit, Tel Aviv, Israel Jul 2026 – Sep 2026
Intern

- Improved GTFS and OpenStreetMap preprocessing so complete intracity bus lines are kept whenever any segment crosses a selected area, using map matching and network edge processing
- Researched transit-agency customers, competitors and market size to shape product positioning and go-to-market

Formula Student Technion, Haifa, Israel Oct 2023 – Apr 2024
High Voltage Engineer

- Co-designed a new high-voltage accumulator for the race car from scratch in SolidWorks, managing files in Kenesto and tasks in monday.com